Biochemistry Lab #8
Assay of Alcohol Dehydrogenase Activity
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Introduction:

Alcohol dehydrogenase (ADH) is an enzyme found in yeast, the livers of animals, and other microorganisms, playing an essential role in ethanol metabolism. ADH catalyzes the oxidation of ethanol to acetaldehyde, utilizing NAD^+ as the oxidizing agent, resulting in NADH as the product.

$$CH_3CH_2OH + NAD^+ -> CH_3CHO + NADH + H^+$$

Spectrophotometry can be used to monitor the formation of NADH. The enzyme activity can be measured as NADH absorbs at 340 nm, while NAD^+ does not. Through tracking the absorbance values, the initial velocity (V_0) of the reaction is measured at the start of the reaction, then as the enzyme converts NAD^+ into NADH, the absorbance increases. The slope found between the initial and final points of the reaction shows the enzymatic activity, or the rate at which an enzyme converts substrate into product. The reaction will be slowed, mimicking a logarithmic shape due to product inhibition or eventually running out of the substrate. Enzyme activity is reported in International Units (IU), where one IU is one micromole of product formed per minute at standard conditions.

After the initial velocity is found at multiple volumes, this relationship is modeled with the Michaelis-Menten equation. In this experiment, ADH activity was measured with a mixture containing ethanol NAD^+ and Tris buffer. The absorbance at 340 nm was used to determine the initial velocity of the reaction and generate a standard curve. This allows for the comparison of enzyme volume to activity and calculates the activity of the ADH stock solution and an unknown enzyme sample.

Results:

The initial velocity of alcohol dehydrogenase (ADH) was measured at 340 nm. Measurements were taken over 10-second intervals to determine the interval with the highest activity for the enzyme. The observed activity decreased over time due to substrate depletion and product buildup as predicted, with the highest level of activity being at the 0-10 second interval, which was then used to estimate further initial velocities.

Table 1: Determination of Initial Velocity

Time Interval (s)	Activity (mIU) for 10 µM ADH
0-10	108
10-20	105
20-30	101
30-40	98
40-50	96

For sample **A,** the enzyme activities of ADH were then measured across eight different increasing volumes. Each assay contained 2.9 mL of cocktail with an increasing amount of enzyme. Activity increased proportionally with the concentration of the enzyme.

Table 2: ADH Initial Velocities for Sample A

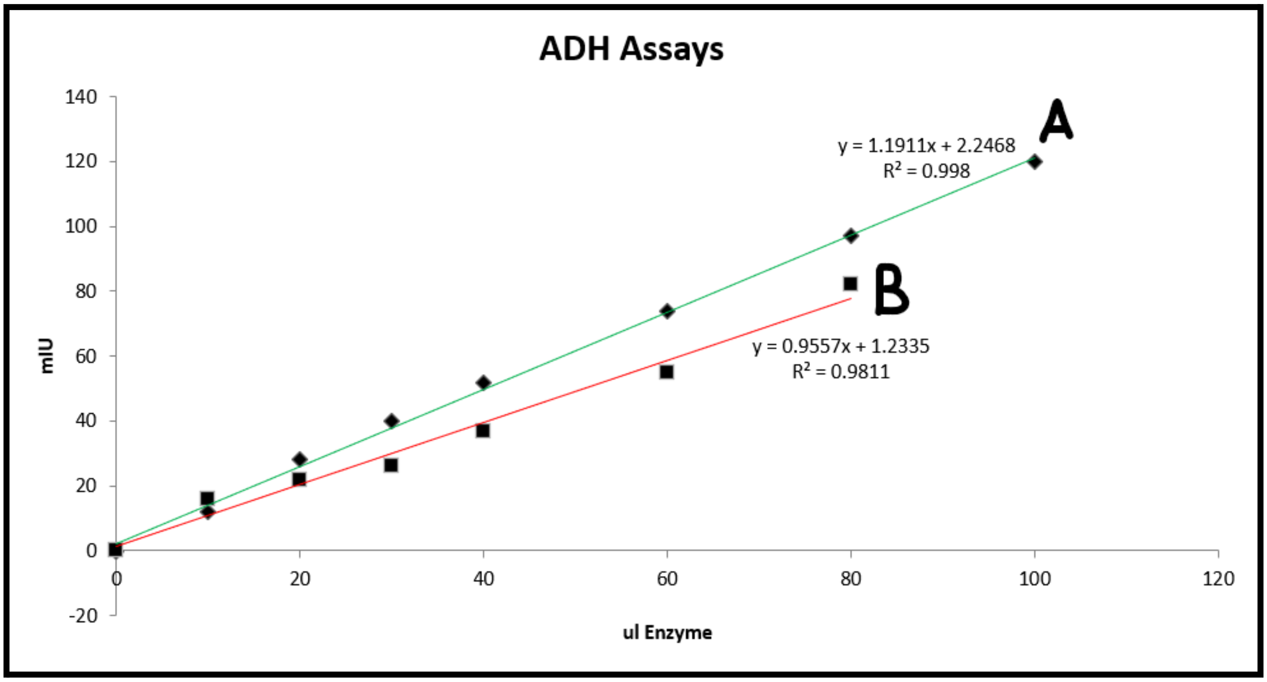
Assay #	H_2O **(µl)**	**ADH (µl)**	V_0 **in mIU for Sample A**
1	100	0	0
2	90	10	12
3	80	20	28
4	70	30	40
5	60	40	52
6	40	60	74
7	20	80	97
8	0	100	120

The activity of the unknown ADH sample **(sample B)** was measured. This was compared to the curve generated from Sample A to determine the enzyme's activity.

Table 3: ADH Initial Velocities of Unknown Sample **(sample B)**

Assay #	H_2O **(µl)**	**ADH (µl)**	V_0 **in mIU for Sample B**
1	100	0	0
2	90	10	16
3	80	20	22
4	70	30	26
5	60	40	37
6	40	60	55
7	20	80	82

Graph 1: Graph of Sample A and the Unknown Sample (B)



Discussion:

The enzymatic activity of alcohol dehydrogenase was successfully measured at 340 nm using absorbance. This activity accurately reflects the formation of NADH. The initial velocity was measured in the 0-10 second window after starting the reaction, as it provides the most accurate estimate for the enzyme's starting initial rate before substrate depletion affects the activity.

The linear plot generated for sample A confirms the proportional relationship between enzyme volume and enzyme activity. The linear graph indicates that the proper experimental conditions were present, the substrate was in excess, and that the measured rates accurately represent the volume of ADH present.

When plotted, sample B was also present in the linear region of sample A. This allows for Sample B’s IU/mL value to be estimated from the slope of Sample A. When comparing the samples, Sample B showed lower activity than Sample A at the equivalent volumes. Indicating that Sample B contains either a lower concentration of enzyme or has a lower specific activity. Both of the samples still displayed a linear relationship between enzyme volume and activity, demonstrating the reliability of using a linear graph to quantify unknown and known ADH samples.

Materials and Methods:

ADH, NAD^+, ethanol, and buffer were used to prepare the cocktail for the reactions. Increasing volumes of ADH were added to the cocktail, and the absorbance at 340 nm was monitored using a Shimadzu-1280 UV-Vis spectrophotometer. Absorbance was recorded at 10-second intervals using a preset protocol that capped at 0.3 absorbance.

Two linear curves were generated for Samples A and B with their absorbances and increasing enzyme volumes. All assays were performed at room temperature with consistent conditions.

References:

Contreras-Zentella, Martha Lucinda, et al. "Ethanol Metabolism in the Liver, the Induction of Oxidant Stress, and the Antioxidant Defense System." *Antioxidants (Basel, Switzerland)*, U.S. National Library of Medicine, 26 June 2022, pmc.ncbi.nlm.nih.gov/articles/PMC9312216/.

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